

High-Level Flowchart

All steps taken in the Decision Tree customer segmentation notebook, with short notes and key outputs.

Learning Day 7 - Decision Trees for Customer Segmentation

Documentation generated from the finalized notebook and project results.

High Level Flowchart of All Steps

This document summarizes all major steps in the notebook. Each flowchart box includes a short action note and the key output created at that stage.

Use this as the big-picture map of the project. The detailed variable names are documented separately in the Variable Flow PDF.

Phase 1: Data Import and Preprocessing

Phase 1: Data Import and Preparation

From raw Kaggle files to model-ready numeric feature matrices.

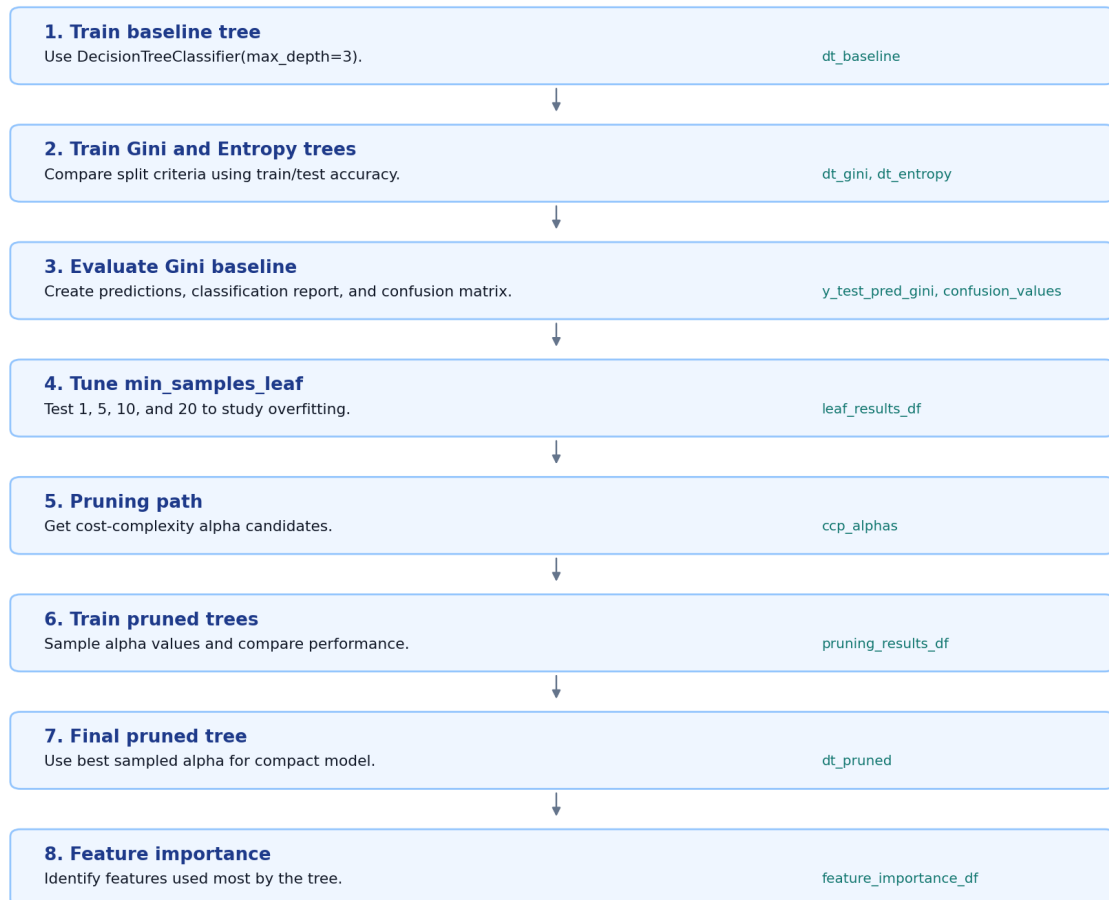


Phase 1 covers Kaggle import, dataset audit, feature/target setup, split, and preprocessing.

Phase 2: Modeling and Tuning

Phase 2: Decision Tree Modeling and Tuning

From baseline model to controlled, interpretable tree tuning.



Phase 2 covers baseline tree, evaluation, leaf-size tuning, pruning, and feature importance.

Phase 3: Expert and Explainability

Phase 3: Expert Comparison and Explainability

From expert oblique comparison to global and local explanation.



Phase 3 covers the custom oblique tree, model comparison, SHAP beeswarm, and waterfall explanation.

One-Page Step Summary

Step	Action	Small note	Main output
1	Download dataset	Use Kaggle API and project-relative data folder	DATA_DIR
2	Load files	Read Train.csv and Test.csv	train_df, test_df
3	Audit data quality	Missing values and duplicates	missing_summary_with_values
4	Check target distribution	Confirm class balance	target_distribution
5	Create X and y	Separate features and target	X, y
6	Train/test split	Stratified split	X_train, X_test, y_train, y_test
7	Preprocess data	Impute + one-hot encode	X_train_processed, X_test_processed
8	Train baseline tree	max_depth=3	dt_baseline
9	Compare criteria	Gini vs Entropy	dt_gini, dt_entropy
10	Evaluate model	Accuracy, report, confusion matrix	y_test_pred_gini, confusion_values
11	Tune leaf size	Study overfitting	leaf_results_df
12	Prune tree	Use ccp_alpha	dt_pruned
13	Feature importance	Explain tree drivers	feature_importance_df
14	Train oblique tree	LinearSVC splitters	custom_oblique_tree
15	Compare models	Standard vs oblique	standard_vs_oblique_df
16	SHAP beeswarm	Global explanation for Segment D	shap_values_segment
17	SHAP waterfall	Local explanation for wrong prediction	misclassified_explanation
18	Final recommendation	Standard Gini tree selected	Final summary